

Chapter 17: A Comprehensive Review of Explainable Artificial Intelligence in Breast Cancer Detection

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Abstract

Early and precise diagnosis is crucial for improving patient outcomes because breast cancer is still one of the top causes of cancer-related mortality among women globally. Although deep learning (DL) and machine learning (ML) approaches have been remarkably successful in detecting breast cancer in recent years, their acceptance in clinical practice has been restricted due to their black-box character. A thorough examination of interpretability methods combined with ML and DL models for breast cancer diagnosis is provided in this article. In addition to more contemporary hybrid and transformer-based methods, it looks at traditional classifiers like Support Vector Machine (SVM), Random Forest (RF), Logistic Regression (LR), Decision Tree (DT), XGBoost, and ensemble models. Explainable artificial intelligence (XAI) techniques, such as SHAP, LIME, Grad-CAM, ELI5, Anchor, and QLattice, which offer both local and global explanations of model predictions, are given special attention. By recognizing important clinical and imaging aspects, interpretability greatly improves transparency, boosting clinician confidence and facilitating well-informed decision-making, as the reviewed research show. Even though several research have achieved classification accuracies higher than 95%, issues with external validation, multimodal data integration, computing efficiency, and standardized interpretability evaluation still exist. The review identifies research gaps, emphasizes current trends, and explores future options for creating clinically reliable, transparent, and accurate breast cancer detection systems. The results offer academics useful direction for creating interpretable AI frameworks that successfully strike a compromise between explainability and prediction performance for practical healthcare applications.

Keywords: Interpretability, breast cancer, machine learning, deep learning, Explainable Artificial Intelligence

1. Introduction

Breast cancer is a condition of abnormal growth of cancer cells in the breast tissue. If untreated, it can spread to the surrounding tissues with a fatality risk. The growth of aberrant breast cells that eventually develop into malignant tumors is what defines breast cancer. These cancer cells have the potential to spread and become deadly. This fatal cancer can develop as a result of a complex combination of individual genetic composition and environmental circumstances. Given the increasing frequency of breast cancer, advancements in cancer diagnosis and therapy are crucial. Machine learning has contributed to reliable medical diagnosis outcomes during the last few decades. Thus, in order to detect breast cancer, this study discusses several machine learning classifiers and patient diagnostic features. Explainable Artificial Intelligence (XAI) approaches added a layer of transparency and interpretability by revealing the underlying variables driving the model predictions.

In order to maximize accuracy and dependability in real-world applications, XAI plays a crucial role in assessing model performance. Beyond performance, it is critical to evaluate how XAI affects user

comprehension and confidence. Ensuring the transparency and interpretability of AI technologies is crucial for building trust between patients and medical professionals as they become more and more incorporated into healthcare. However, integrating different XAI methodologies poses a number of difficulties, especially when it comes to comparative analysis and generalization across numerous circumstances. Standardized evaluation metrics are essential because they offer dependable and consistent assessments of efficacy in clinical contexts. These metrics should measure the quality of the explanations in terms of relevance, completeness, and consistency in addition to the accuracy and effectiveness of XAI applications. Moreover, the integration of XAI into clinical workflows can advance personalized medicine.

Some of the Explainable AI (XAI) techniques discussed in the related study are presented below:

SHAP (SHapley Additive exPlanations) : SHAP measures each feature's contribution to a model's prediction using an explainability technique based on game theory [1]. It offers explanations that are both local (particular prediction) and global (overall model). Because SHAP guarantees consistency and interpretability, it is appropriate for clinical decision assistance in the diagnosis of breast cancer. It is widely applied in identification of significant clinical and imaging characteristics.

LIME (Local Interpretable Model-agnostic Explanations): LIME explains individual predictions by approximating the complex model with a straightforward, comprehensible model surrounding a particular instance [2]. It perturbs the input data and examines the changes in prediction. Both deep learning and machine learning models can be explained using LIME, which is model-agnostic. It is frequently used to interpret the results of breast cancer categorization.

Grad-CAM (Gradient-weighted Class Activation Mapping): A visualization method created especially for convolutional neural networks (CNNs) is called Grad-CAM. It creates heatmaps that show the areas of the image that are most important to the prediction [3,4]. Grad-CAM assists physicians in determining whether the model concentrates on suspicious lesions or tissue anomalies in breast cancer imaging. It enhances transparency in medical image analysis.

ELI5 (Explain Like I'm Five): An open-source Python package called ELI5 uses feature importance, decision rules, and weight display to explain machine learning model predictions [5]. Text classifiers, tree-based models, and linear models are all supported. ELI5 helps researchers validate significant predictive characteristics in healthcare datasets and makes model behavior easier to comprehend.

Anchor: Anchor finds criteria (anchors) that ensure a model's prediction remains the same in order to produce high-precision, rule-based explanations [6]. Clinicians can easily comprehend these IF-THEN rules since they are intuitive. Anchor is useful for reliable clinical decision-making since it offers explanations with quantifiable precision and coverage.

QLattice: QLattice is an interpretable symbolic machine learning framework that finds basic mathematical relationships between predictions and input variables automatically [7]. It naturally generates transparent predictive models rather than serving as a post-hoc explanation technique. QLattice allows doctors to comprehend feature interactions in breast cancer research while still achieving competitive predicting results.

2. Related Study

An interpretable machine learning-based breast cancer classification using XGBoost, SHAP, and LIME: This research [8] focuses on improving breast cancer classification accuracy through machine learning (ML) methodologies, emphasizing interpretability. The study utilized the chisquare method to

enhance model testing performance by pinpointing the most significant features for further analysis. The study also improved data quality by identifying and removing outliers, thus minimizing the influence of data irregularities on the performance of the models. For classification, the study evaluated six different ML algorithms—namely extreme gradient boosting (XGBoost), decision tree (DT), AdaBoost (AB), support vector machine (SVM), gradient boosting (GB), and K-nearest neighbors (KNN)—each applied to distinguish between the two variants of breast cancer. Among these, the XGBoost classifier emerged as the most accurate, achieving an impressive 99.30% accuracy rate. Additionally, to improve the interpretability of the suggested model and provide important insights into the model's decision-making process, the study included shapley additive explanations (SHAP) and local interpretable model-agnostic explanations (LIME) techniques. By using these interpretability methodologies, considerable insights into the predictive aspects influencing healthcare outcomes were obtained, guaranteeing the transparency and dependability of the categorization strategy.

Towards an interpretable breast cancer detection and diagnosis system: This paper [9] proposes a five-step automated and comprehensible system for breast cancer detection and diagnosis. Following a thorough pre-processing and unsupervised segmentation, we examine three feature selection strategies and four textural and shape-based feature extraction methodologies. We use the Decision Tree algorithm for benign/malignant classification to make interpretation easier, and we experiment with pre-pruning, post-pruning, and ensemble-based (Random Forest classifier) techniques to prevent overfitting. When evaluated on photos from the mini-MIAS dataset, our system achieves a maximum accuracy of 95% and 100% precision and specificity. It also allows its users to examine each step. Because each processing step is explained visually, the suggested system stands out for its interpretability. It is quicker and easier to train than Artificial Neural Networks, which is important in the healthcare industry because datasets are frequently small. The system also incorporates the threshold-based GrowCut algorithm. The user can confirm each stage of the segmentation process by employing an iterative segmentation strategy.

Detection of breast cancer using machine learning and explainable artificial intelligence: This study [10] predicted breast cancer diagnosis using explainable AI and machine learning techniques. With an F1-score of 84%, random forest produced the best results among the methods. The stacked ensemble model achieved an F1-score performance of 83% by combining the strengths of many models. In order to interpret the results, the study focused on the outcomes of explainers such SHAP (SHapley Additive exPlanations), LIME (Local Interpretable Model-agnostic Explanations), ELI5 (Explain Like I'm Five), Anchor and QLattice (Quantum Lattice). In the medical field, interpretable algorithms can help practitioners to predict breast cancer, decrease diagnostic errors, and enhance clinical decision-making.

Breast cancer diagnosis with Explainable Artificial Intelligence (XAI): Uncovering Strengths and Biases: The application of XAI techniques in the detection and diagnosis of breast cancer is the main topic of this survey [11], with a focus on how these techniques might improve model transparency and clinical decision-making. Convolutional Neural Networks (CNNs) compatibility with visual explanation techniques and tree-based models with feature importance evaluations are just two examples of the inherent biases in the most current machine learning models toward particular XAI methodologies that are discussed in the article. The article concludes by discussing the challenges associated with implementing XAI technologies in clinical settings and the importance of establishing common metrics for evaluating their effectiveness.

Interpretable Machine Learning Approach for Breast Cancer Classification: This study [12] assessed four machine learning models for breast cancer classification: Logistic Regression, Decision

Trees, Random Forest, and CatBoost using three datasets WBC, WDBC, and Coimbra. Logistic Regression performed the best, achieving up to 97% accuracy on WDBC and being quick enough for real-time application. Although Random Forest was slower, its accuracy was comparable. Decision Trees and CatBoost did not perform well, especially on the Coimbra dataset. By emphasizing crucial characteristics including tumor size (WDBC), cell thickness (WBC), and BMI (Coimbra), the study employed LIME to explain model decisions, increasing the results' dependability for experts. This work uses more datasets and provides clearer insights, improving reliability and usefulness in compared to previous studies such as Wei et al. [13], which achieved 94% accuracy but lacked explanations. While resource constraints limited the use of deep learning models in this study, lightweight model architectures such as MobileNet or EfficientNet might prove useful in future research. These models are designed for low computing demand and can handle histopathological images or clinical feature embeddings, possibly improving performance on datasets such as Coimbra when combined with transfer learning. Using larger datasets like TCGA, future research should work with physicians to increase accuracy and confidence in low-resource environments. Additionally, integrating tools like SHAP to provide global feature importance alongside LIME's local explanations would improve interpretability and clinical alignment.

Development and validation of an Interpretable Machine learning model for Discriminating between benign and malignant breast cancer: It is still difficult to distinguish between breast lesions. In this study [14], the author created a model that emphasizes low cost and scalability utilizing common, easily accessible clinical and laboratory variables (Age, TT, APTT, CEA, Ca). AUC, Brier score, calibration curve, and decision curve analysis (DCA) for clinical net benefit and performance are reported by the model; external validation is used to improve robustness. For interpretability, SHAP was employed, and an online application was made available to enhance clinical usefulness and transparency.

An interpretable Multi-Dataset learning framework for breast cancer prediction using clinical and biomedical tabular data: In addition to traditional machine learning and deep learning techniques, this study [15] proposes a thorough multimodal framework for automated breast cancer diagnosis that incorporates a variety of biological data. Transformer-based BERT achieves only 92.98% accuracy on tabular data, but conventional models like SVM and Random Forest get 94–97% accuracy and RNN-based models reach 98.25%. This underscores the need for specialized architectures. These constraints are addressed by our suggested FT-Transformer-Attention-LSTM-SVM hybrid model, which uses margin-optimized classification, sequential attention modeling, and self-attention feature encoding. On the WBCO dataset, it outperforms baseline approaches by achieving 99.90% test accuracy, 99.56% mean cross-validation accuracy, and 98.50% external validation accuracy. Its robustness is confirmed by statistical analysis ($t = 14.72$, $p < 0.0001$; Wilcoxon $p < 0.0001$; Cohen's $d = 4.65$), with only three misclassifications over all cross-validation folds. Clinically meaningful interpretability is demonstrated by SHAP and LIME analyses, and real-time deployment is supported by the lightweight architecture (<1 ms inference time, 530 K parameters). With future possibilities for multi-class staging, genomic integration, and prospective clinical validation, our findings demonstrate that the suggested model is a reliable, effective, and comprehensible method for predicting breast cancer using tabular data.

An interpretable hybrid AI model for breast fine needle aspiration cytology image classification: This study [16] presents a hybrid technique for breast cancer diagnosis based on FNAC images that incorporates ML classifiers and DL algorithms as feature extractors. Two hybrid approaches are investigated to differentiate between benign and malignant FNAC images. In the first approach, a single

pre-trained network is combined with an ML classifier. In the second approach, two previously trained models' feature vectors are concatenated before being fed into an ML classifier. The hybrid model, which combines DenseNet-121 and MobileNet-V2 with an SVM classifier, gets the highest testing accuracy of 98.26% using the Adam optimizer with a learning rate of 0.001. An explainable AI (XAI) method called Grad-CAM is used to for the proposed model explainability. Expert pathologists acquired 95% positive clinical validation for Grad-CAM, demonstrating the model's reliability and interpretability.

3. Discussions on Related Study

The following discussion points offer a critical evaluation of the current research, highlighting significant limits, current trends, and possible future paths based on the previously evaluated studies.

3.1 High Classification Accuracy but Dependency on Dataset

Using traditional machine learning methods like XGBoost, Random Forest, Logistic Regression, and SVM, the majority of research achieved strong classification performance with accuracies ranging from 95% to 99.30%. These findings, however, were mostly based on benchmark datasets (such as WDBC, WBC, Coimbra, and mini-MIAS), which restricted the models' applicability in a variety of clinical settings.

3.2 The significance of Data Preprocessing and Feature Selection

By identifying the most informative features, feature selection methods like the Chi-square method greatly enhanced model performance. In a similar vein, preprocessing techniques including feature extraction, segmentation, and outlier elimination improved data quality and decreased noise, resulting in more accurate forecasts.

3.3 Growing Adoption of Explainable Artificial Intelligence (XAI)

To increase transparency, recent research has increasingly included explainability techniques like **SHAP**, **LIME**, **ELI5**, **Anchor**, and **QLattice**. These techniques boosted clinician confidence and supported well-informed medical decision-making by identifying significant features that contributed to predictions.

3.4 Preference for Interpretable Machine Learning Models

A number of researchers favored models that are easily understood, like Decision Trees and Logistic Regression, particularly in healthcare applications where transparency is crucial. These models make clinical interpretation and validation simpler, even though they occasionally produce somewhat lower accuracy than complex ensembles.

3.5 Trade-off Between Accuracy and Interpretability

Although ensemble models like XGBoost, Random Forest, and CatBoost needed post-hoc interpretability techniques like SHAP and LIME, they typically produced better prediction performance. This emphasizes the continuous difficulty in striking a balance between clinical interpretability and model complexity.

3.6 Limited Use of Deep Learning in Structured Clinical Data

Despite deep learning models' impressive performance in medical imaging, their use in structured clinical datasets is still restricted because of smaller sample numbers and processing demands. Lightweight architectures and transfer learning have been proposed as viable future prospects in a number of research.

3.7 Need for Robust Validation

Internal validation using benchmark datasets was the main method used in several research. Few research incorporated **external validation**, calibration analysis, or decision curve analysis, which are essential for evaluating the resilience of the model and its practical clinical relevance.

3.8 Interpretability Beyond Feature Importance

Local decision explanations or feature importance is the primary focus of existing explainability techniques. Few research provided thorough insights into decision consistency across various patient groups, uncertainty estimation, or feature relationships.

3.9 Clinical Applicability is still a Major Obstacle

Although a number of models showed outstanding predictive performance, practical implementation elements including online clinical decision support systems, integration into hospital workflows, computational efficiency, and usability by healthcare professionals received little attention.

3.10 The Development of Multimodal and Hybrid Learning Frameworks

The focus of recent research has switched to hybrid architectures that combine transformer-based models, deep learning, and conventional machine learning. These methods provide significant gains in classification accuracy and robustness by utilizing the complimentary qualities of several algorithms.

4. Future Research Directions

Explainable artificial intelligence (XAI) for breast cancer screening has made great strides, but there are still a number of obstacles to overcome before these models are widely used in clinical settings. Improving interpretable AI systems' scalability, clinical applicability, and dependability should be the main goals of future research. The following directions represent key areas for future investigation.

4.1 Validation on Larger and Multi-Institutional Datasets

The majority of interpretability studies that are currently available have been assessed using relatively small, single-center datasets, which restricts their applicability to a variety of patient demographics and imaging settings. Future studies should use extensive, multi-institutional datasets gathered from various hospitals, geographical areas, and imaging technologies to validate interpretable models. Such validation would improve the robustness of explanations by taking into consideration differences in image collecting procedures, disease prevalence, demographics, and clinical practices. Additionally, multi-center assessments would boost confidence in implementing AI systems in diverse healthcare settings and reduce the risk of model overfitting.

4.2 Combining Local and Global Explainability Techniques

The majority of explainable AI techniques offer either global explanations that characterize the model's general behavior or local explanations that support specific predictions. But depending only on one viewpoint could not provide us enough information about how the model makes decisions. Hybrid explainability frameworks that include local and global interpretability techniques should be the main focus of future research. While global methods like feature importance analysis, surrogate models, and

rule extraction can show the overall decision logic, local methods like Grad-CAM, SHAP, and LIME can explain patient-specific predictions. Combining these complementary viewpoints would increase openness, make error analysis easier, and boost doctor confidence in AI-assisted diagnosis.

4.3 Improving Computational Efficiency for Real-Time Deployment

Many state-of-the-art explainability techniques require substantial computational resources, making them difficult to integrate into time-sensitive clinical workflows. Future research should focus on creating computationally efficient, lightweight explanation algorithms that produce high-quality explanations with minimal latency. Model compression, hardware acceleration, effective feature attribution strategies, and model optimization techniques can all greatly lower computational cost without sacrificing explanation fidelity. Real-time interpretability will be particularly important for computer-aided diagnosis systems functioning in busy clinical environments where quick decision support is crucial.

4.4 Enhancing External Validation and Clinical Usability

The clinical use of many explainability techniques has not been adequately assessed, despite the fact that they provide visually intuitive heatmaps and feature importance scores. Future research should include extensive external validation using independent datasets and prospective clinical trials including radiologists, pathologists, oncologists, and cytologists. A human-centered review should assess whether explanations enhance clinical decision-making, decrease interpretation time, increase diagnostic confidence, and facilitate error detection easier. The practical usability of interpretable AI systems can also be improved by user-friendly interfaces that let physicians interact with explanations, modify visualization levels, and investigate model thinking.

4.5 Developing Trustworthy AI Systems that Balance Predictive Performance with Interpretability

Future research should prioritize the development of reliable AI systems that achieve high diagnostic accuracy while retaining transparency and interpretability. Rather than treating predictive performance and explainability as competing objectives, researchers should investigate methods that optimize both simultaneously. Alongside interpretable decision-making, such systems should include uncertainty estimation, fairness evaluation, robustness against adversarial perturbations, bias identification, and privacy-preserving learning. Additionally, explanations must to be coherent, true to the underlying model, and clinically significant so that medical experts may confidently validate AI suggestions. Promoting the safe and responsible use of AI in breast cancer detection will require the establishment of standardized evaluation frameworks that simultaneously measure accuracy, interpretability, reliability, and clinical value.

5. Conclusions

The reviewed literature demonstrates significant advancements in the use of explainable AI and machine learning for breast cancer diagnosis. Conventional machine learning models continue to offer great predictive performance, while explainability methods like SHAP and LIME increase transparency and clinician trust. However, challenges remain regarding multimodal data integration, external validation, dataset diversity, and application in actual clinical settings. A promising approach is represented by recent hybrid architectures that offer both improved resilience and greater predicted accuracy by combining transformer-based feature learning with traditional classifiers.

The adoption of explainable AI in healthcare will be greatly increased by validating models on sizable multi-institutional datasets, integrating multimodal clinical and imaging data, combining local and global

explainability techniques, increasing computational efficiency for real-time applications, strengthening external clinical validation, and creating reliable AI frameworks. These advancements will enable clinicians to better understand model predictions, improve diagnostic confidence, enhance individualized treatment choices, and ultimately facilitate the responsible integration of AI into routine breast cancer screening and diagnosis.

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